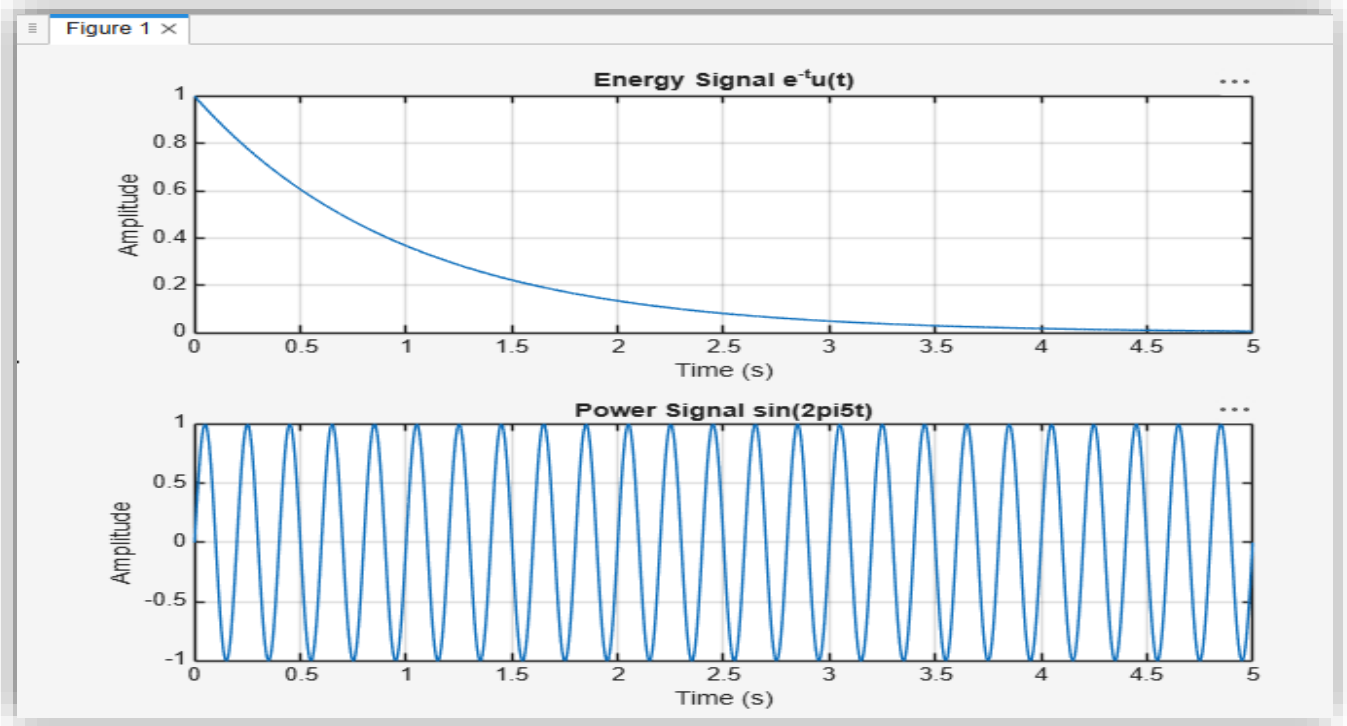
Experiment 3

Energy and Power Analysis of Signals

/MATLAB Drive/Experiment_3.m

```
1   clc;
2   clear;
3   close all;
4   dt = 0.001;
5   t = 0:dt:5;
6   xE = exp(-t);
7   xP = sin(2*pi*5*t);
8   E_num = trapz(t,abs(xE).^2);
9   T0 = 1/5;
10  idx = t <= 2;
11  P_num = (1/(t(find(idx,1,'last'))-t(1))) * ...
12      trapz(t(idx),abs(xP(idx)).^2);
13  figure;
14  subplot(2,1,1);
15  plot(t,xE); grid on;
16  title('Energy Signal e^{-t}u(t)');
17  xlabel('Time (s)'); ylabel('Amplitude');
18  subplot(2,1,2);
19  plot(t,xP); grid on;
20  title('Power Signal sin(2pi5t)');
21  xlabel('Time (s)'); ylabel('Amplitude');
22  fprintf('Numerical energy = %g\n',E_num);
23  fprintf('Numerical average power = %g\n',P_num);
```

Code



Output